

LA RONGE, SK, Canada

WMO: 715860

Lat: 55.1464N

Lon: 105.2703W

Elev: 379

StdP: 96.86

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.8	-33.3	-40.1	0.1	-35.4	-37.5	0.1	-33.1	9.3	-14.1	8.3	-15.5	1.2	280	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	27.8	17.7	26.1	16.9	24.4	16.1	19.1	25.3	18.2	23.9	17.3	22.5	4.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.9	12.6	21.3	16.0	11.9	20.4	15.2	11.3	19.6	56.1	25.4	53.0	24.2	50.1	22.4	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.8	6.8	-39.4	31.7	2.8	1.8	-41.5	33.0	-43.1	34.0	-44.7	35.1	-46.8	36.4		
			-39.5	21.1	2.8	0.8	-41.5	21.6	-43.2	22.1	-44.8	22.6	-46.8	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-18.3	-15.8	-8.5	0.9	8.4	14.7	17.7	16.3	10.6	2.9	-7.3	-15.7
	DBStd	14.15	8.94	7.89	8.25	6.28	4.88	3.67	2.99	3.54	4.49	4.61	6.70	8.56
	HDD10.0	4134	877	723	573	275	91	6	0	2	47	225	518	798
	HDD18.3	6543	1136	956	832	522	310	119	48	83	234	479	768	1056
	CDD10.0	691	0	0	0	4	39	146	238	196	64	4	0	0
	CDD18.3	58	0	0	0	0	1	9	28	19	2	0	0	0
	CDH23.3	596	0	0	0	1	26	104	262	177	25	0	0	0
	CDH26.7	109	0	0	0	0	4	18	48	36	3	0	0	0
Wind	WSAvg	3.0	2.7	2.8	3.1	3.3	3.4	3.2	3.0	3.0	3.2	3.3	3.0	2.6
Precipitation	PrecAvg	501	19	16	20	31	44	77	86	63	58	39	29	24
	PrecMax	757	39	36	59	76	129	298	201	151	149	74	52	45
	PrecMin	321	6	2	4	1	2	11	40	9	15	10	3	8
	PrecStd	89	9	7	13	20	32	48	37	33	34	15	13	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	5.3	11.0	20.5	26.5	28.9	30.2	30.0	26.1	18.7	7.9	3.1
		MCWB	1.8	1.7	4.6	10.2	14.1	17.2	19.1	19.3	16.2	11.9	4.0	0.7
	2%	DB	0.4	1.3	7.6	16.7	22.9	26.1	27.9	27.3	23.2	14.7	5.3	0.2
		MCWB	-1.0	-0.7	2.8	8.5	12.6	16.1	18.2	17.9	15.3	9.4	2.5	-1.0
	5%	DB	-2.7	-1.4	5.0	13.4	20.5	24.1	26.2	25.3	20.4	12.0	2.9	-2.0
		MCWB	-3.6	-2.8	1.4	6.5	11.4	15.0	17.4	17.2	14.0	7.9	1.1	-3.0
	10%	DB	-5.3	-3.8	2.5	10.4	18.0	22.2	24.5	23.2	18.0	9.6	1.0	-4.3
		MCWB	-6.0	-4.7	0.1	4.9	10.1	14.2	16.7	16.3	12.7	6.6	-0.2	-5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.2	5.1	10.9	15.3	19.3	20.7	20.4	17.6	12.6	4.6	1.4
		MCDB	4.0	5.1	10.4	19.6	23.7	25.7	27.4	27.4	23.2	17.9	7.2	2.8
	2%	WB	-1.0	-0.6	3.1	8.5	13.5	17.6	19.3	18.9	15.9	10.0	2.8	-1.0
		MCDB	0.4	1.2	7.0	15.8	20.7	23.2	25.5	25.2	21.5	13.4	4.7	0.2
	5%	WB	-3.6	-2.7	1.6	6.8	12.1	16.5	18.4	18.0	14.6	8.2	1.1	-2.9
		MCDB	-2.7	-1.4	4.6	12.9	19.3	21.8	24.2	23.7	19.7	11.5	2.7	-2.0
	10%	WB	-5.9	-4.9	0.2	5.1	10.6	15.3	17.6	17.0	13.2	6.7	-0.1	-4.9
		MCDB	-5.2	-3.8	2.5	10.0	16.9	20.3	22.9	21.9	17.3	9.3	1.1	-4.3
Mean Daily Temperature Range	5% DB	MDBR	9.1	11.1	12.0	11.8	12.3	10.9	10.9	10.6	9.9	7.6	6.9	8.1
		MCDBR	10.9	12.0	13.2	15.9	16.9	14.4	13.8	13.8	14.0	12.0	8.6	9.4
		MCWBR	9.7	10.1	9.1	9.3	8.6	6.6	6.2	6.5	7.5	7.5	6.3	8.4
	5% WB	MCDBR	10.7	11.7	12.0	15.3	15.0	11.4	12.2	12.2	13.2	11.3	7.8	9.1
		MCWBR	9.6	9.9	8.5	9.1	7.9	5.8	5.9	6.2	7.5	7.3	6.1	8.2
Clear-Sky Solar Irradiance	taub	0.238	0.245	0.272	0.314	0.336	0.365	0.376	0.357	0.312	0.288	0.243	0.214	
	taud	2.288	2.348	2.363	2.453	2.474	2.425	2.400	2.477	2.590	2.577	2.434	2.308	
	Ebn at Noon	726	865	914	916	909	881	862	860	865	810	731	687	
	Edh at Noon	52	71	90	97	101	107	108	94	74	59	45	40	
All-Sky Solar Radiation	RadAvg	0.80	1.84	3.38	4.88	5.53	5.59	5.49	4.58	3.11	1.67	0.75	0.52	
	RadStd	0.08	0.11	0.26	0.33	0.52	0.38	0.34	0.40	0.35	0.25	0.11	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page